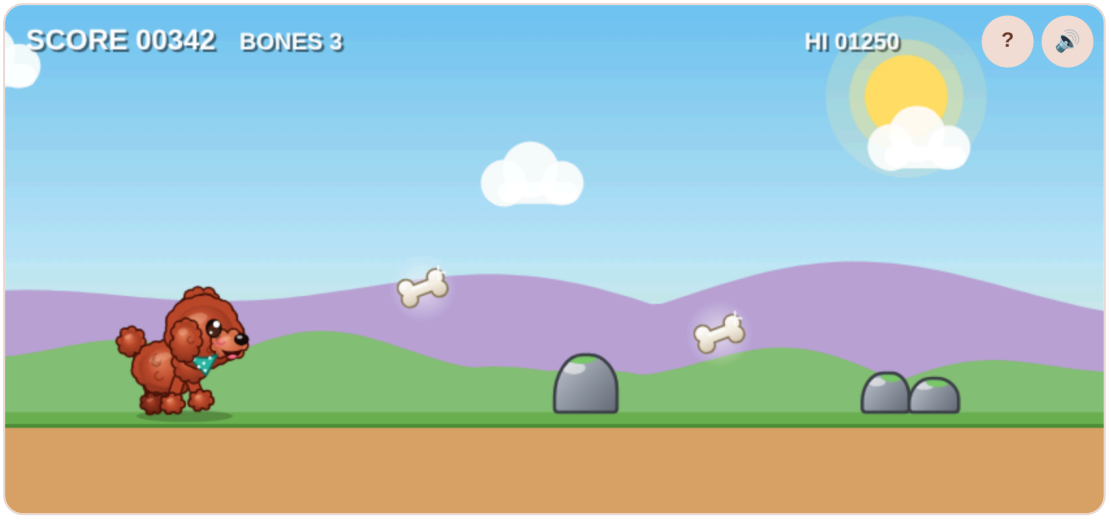


Rocky Run

Release Guide: Android APK & Website



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Created by Jovita

Flutter 3.47.7 (stable) · Windows 11 · Android SDK 36

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1. Release at a glance

Rocky Run ships from one codebase to two places: an **APK** file people install on Android phones, and a **website** that runs in any browser, including phones.

	Android APK	Website
Needs	`flutter doctor` → **[√]** Android toolchain	Only Chrome (already working)
Build command	`flutter build apk --release`	`flutter build web --release`
Output	`build\app\outputs\flutter-apk\app-release.apk`	Folder `build\web`
Where it goes	Phone (USB / file), Google Drive, GitHub Releases, Play Store	Netlify or GitHub Pages (free)
Players get	Home-screen app with the Rocky icon, works offline	A link, nothing to install
Time	First build 3–5 min, then ~1 min	~1 min build + 1 min upload

2. Before every release

1. **Check the tools.** Run `flutter doctor`. For the APK, **Android toolchain** must be **[√]**.
2. **Raise the version** in `pubspec.yaml`. The format is `name+build`:

   ```
   version: 1.0.0+1   # first release
   version: 1.0.1+2   # next release: bump the name, and ALWAYS +1 the build number
   ```

   Android refuses to install an update whose build number is not higher than the installed one.
3. **Test** on Chrome and on the emulator or a phone: start screen, How to Play, a full run, losing all 3 hearts, Settings (music, sound, language), Run again / Title / Exit.
4. **Clean build** (removes stale icons and caches):

```
cd C:\rocky_run
flutter clean
flutter pub get
```

3. Part A · Android APK

A1 · App identity (one time)

Three things identify the app on a phone. Two are already done for Rocky Run.

Item	Where	Status
App name "Rocky Run"	`android\app\src\main\AndroidManifest.xml` → `android:label`	**Done**
Launcher icon (Rocky's face)	`android\app\src\main\res\mipmap-*`	**Done**
Application ID	`android\app\build.gradle.kts` → `applicationId`	Still `com.example.rocky_run`

The application ID is the app's permanent unique name. `com.example…` is fine for personal testing, but the Play Store rejects it. Change it once, before you share the app:

```
// android\app\build.gradle.kts  (inside defaultConfig)
applicationId = "com.jovita.rockyrun"
```

> **Choose carefully:** after the app is published or installed by others, the ID can never change. A different ID is treated as a different app. Leave `namespace` as it is.

A2 · Quick test APK (no setup)

```
flutter build apk --release
```

This APK is signed with Flutter's **debug key**. It installs and runs normally, which is perfect for testing on your own phone. Use A3 before you give the app to other people or upload it to the Play Store.

A3 · Release signing key (one time)

Every Android app is signed. Updates must be signed with the **same key**, or phones refuse to install them. Create your own key once and keep it forever.

1. Create the key (PowerShell). It asks for a password and your name; remember the password.

```
& "C:\Program Files\Android\Android Studio\jbr\bin\keytool.exe" -genkey -v `
  -keystore "$env:USERPROFILE\rocky-run-upload.jks" `
  -keyalg RSA -keysize 2048 -validity 10000 -alias upload
```

2. Create `C:\rocky_run\android\key.properties` with your values:

```
storePassword=YOUR_PASSWORD
keyPassword=YOUR_PASSWORD
keyAlias=upload
storeFile=C:/Users/blanc/rocky-run-upload.jks
```

3. Edit `android\app\build.gradle.kts`. Add the two imports and the property loader at the **very top** of the file:

```
import java.io.FileInputStream
import java.util.Properties

val keystoreProperties = Properties()
val keystorePropertiesFile = rootProject.file("key.properties")
if (keystorePropertiesFile.exists()) {
    keystoreProperties.load(FileInputStream(keystorePropertiesFile))
}
```

Then, inside `android { … }`, add `signingConfigs` and change the release `signingConfig`:

```
android {
    // ... existing namespace, compileSdk, defaultConfig ...

    signingConfigs {
        create("release") {
            keyAlias = keystoreProperties["keyAlias"] as String
            keyPassword = keystoreProperties["keyPassword"] as String
            storeFile = keystoreProperties["storeFile"]?.let { file(it) }
            storePassword = keystoreProperties["storePassword"] as String
        }
    }

    buildTypes {
        release {
            signingConfig = signingConfigs.getByName("release")   // was "debug"
        }
    }
}
```

Protect the key. Back up `rocky-run-upload.jks` and its password (e.g. a password manager plus a private cloud folder). If it is lost, you can never publish an update to the same app. Never commit it: add these two lines to `C:\rocky_run\.gitignore`:

```
android/key.properties
*.jks
```

A4 · Build the release

Goal	Command	Output
One APK for every phone (simplest)	`flutter build apk --release`	`build\app\outputs\flutter-apk\app-release.apk`

Smaller APKs, one per phone type	`flutter build apk --release --split-per-abi`	`app-arm64-v8a-release.apk` fits almost all modern phones
Google Play Store upload	`flutter build appbundle --release`	`build\app\outputs\bundle\release\app-release.aab`

A5 · Install and test on a phone

- **USB:** on the phone, open **Settings → About phone**, tap **Build number** 7 times, then **Developer options → USB debugging** on. Plug in, allow the prompt, run `flutter install`.
- **File:** send the APK to the phone (Google Drive, WhatsApp, USB copy), open it, allow **Install unknown apps** for that app when asked.

Phone test checklist: the icon and name show on the home screen · How to Play opens on launch · tap to jump, tap again to double jump · music keeps playing through jumps and bones · the phone's Back button returns to the title during a run · portrait and landscape both look right · EXIT closes the app.

A6 · Share or publish

Channel	How	Good for
Google Drive link	Upload the APK, share the link	Friends and testers
GitHub Releases	Repository → **Releases → Draft a new release** → tag `v1.0.0` → attach the APK	Portfolio, version history
Google Play Store	Google Play Console developer account (one-time registration fee) → create app → upload the `.aab` → store listing, screenshots, content rating, privacy policy → review	Public release

> Play Store uploads need the release key from A3 and a unique application ID from A1. Use `assets\icon\rocky_icon_1024.png` as the source for the 512 × 512 store icon.

4. Part B · Website

B1 · Build and test locally

```
cd C:\rocky_run
flutter build web --release
```

The whole website is now in `build\web`. Opening `index.html` by double-click does **not** work (browsers block it). Test with a small local server instead:

```
cd build\web
python -m http.server 8000      # then open http://localhost:8000
```

Or run it straight from VS Code with the **Chrome** device and F5.

B2 · Publish on Netlify recommended

1. Open **app.netlify.com/drop** and sign up free (Google or GitHub login).
2. Drag the `build\web` folder onto the page. A link like `https://random-name-123.netlify.app` appears in seconds.
3. Give it a proper name: **Site configuration → Change site name** → `rocky-run` → `https://rocky-run.netlify.app` (if free).
4. Open the link on PC and phone and run the test checklist from A5 (minus the Back button and EXIT).

B3 · Publish on GitHub Pages alternative

Uses your GitHub account **vita1288**. Create an empty public repository named `rocky_run` first. Then build with the repository name as the base path. The slashes on both sides are required:

```
flutter build web --release --base-href /rocky_run/
cd build\web
git init
git add .
git commit -m "Rocky Run web 1.0.0"
git branch -M gh-pages
git remote add origin https://github.com/vita1288/rocky_run.git
git push -f origin gh-pages
```

On GitHub: repository → **Settings → Pages → Branch: gh-pages / (root) → Save**. After 1–2 minutes the game is live at `https://vita1288.github.io/rocky_run/`.

B4 · Updating the website

1. Raise the version (section 2), then `flutter build web --release` (add `--base-href /rocky_run/` for GitHub Pages).
2. **Netlify:** your site → **Deploys** → drag the new `build\web` folder in. **GitHub Pages:** repeat the B3 git commands.
3. Browsers cache the old version. Check the update with **Ctrl + F5** (PC) or a private tab.

> **How the web version behaves:** music starts after the first tap or click (browser rule) · the EXIT button is hidden because a page cannot close its own tab · on phones it shows tap wording instead of keyboard keys · the first load downloads a few MB, later loads are faster.

5. Release checklist

Android APK	Website
☐ Version and build number raised in `pubspec.yaml` ☐ `flutter doctor`: Android toolchain [√] ☐ Application ID set (not `com.example`) ☐ Release key created and backed up ☐ `key.properties` + `.jks` in `.gitignore` ☐ `flutter clean` → `flutter build apk --release` ☐ Installed on a real phone, phone checklist passed ☐ APK shared / Release created / AAB uploaded	☐ Version raised in `pubspec.yaml` ☐ Tested in Chrome (PC) and a phone browser ☐ `flutter build web --release` (+ base-href for GitHub Pages) ☐ Tested locally at `http://localhost:8000` ☐ Uploaded to Netlify / pushed to `gh-pages` ☐ Live link opened with Ctrl + F5; game loads, music plays after tap ☐ Link shared

6. Troubleshooting

Problem	Cause	Fix
`flutter build apk` fails: Android toolchain / cmdline-tools / SDK 36	Android setup incomplete	Follow the Flutter setup guide, Step 2; then `flutter doctor --android-licenses`
`JAVA_HOME is not set` or Gradle cannot find Java	Java not configured	`flutter config --jdk-dir "C:\Program Files\Android\Android Studio\jbr"`
Build error: `keystoreProperties["keyAlias"] must not be null`	`key.properties` missing or misnamed	File must be `android\key.properties` with exactly the four keys from A3
Phone says “App not installed”	An older copy signed with a different key, or a lower build number	Uninstall the old Rocky Run first, or raise the build number (`+2` , `+3` ...)
“Blocked by Play Protect” when installing	Phone protection for apps from outside the Play Store	Tap **More details → Install anyway**
Old icon still shows	Cached build or launcher cache	`flutter clean` , rebuild, uninstall and reinstall
Website is a blank white page on GitHub Pages	Wrong base path	Rebuild with `--base-href /rocky_run/` (must match the repo name, slashes included)
Website shows the old version	Browser cache	Ctrl + F5, or a private window
No sound on the website	Browsers block sound until the first interaction	Tap or click once; check the phone is not on silent

`index.html` opened by double-click shows nothing	Browsers block `file://` apps	Use `python -m http.server` (B1) or the live link

7. Command reference

Command	Purpose
`flutter doctor`	Check the setup before building
`flutter clean` → `flutter pub get`	Fresh build (stale icons, odd errors)
`flutter build apk --release`	Universal APK
`flutter build apk --release --split-per-abi`	Smaller APK per phone type
`flutter build appbundle --release`	Play Store bundle (.aab)
`flutter install`	Install the built APK on a USB phone
`flutter build web --release`	Website into `build\web`
`flutter build web --release --base-href /rocky_run/`	Website for GitHub Pages
`python -m http.server 8000` (in `build\web`)	Test the website locally

Rocky Run · Release Guide v1.0 · Created by Jovita · Commands assume the project is at `C:\rocky_run` and the Windows user is `blanc`; adjust paths if yours differ.